STAD29 / STA 1007 assignment 5

Due Tuesday Mar 12 at 11:59pm on Quercus

Packages for this one:

```
library(MASS)
library(car) # for Manova later

## Loading required package: carData

library(tidyverse)

## -- Attaching packages ---------------------------------- tidyverse 1.2.1 --
## v ggplot2 3.1.0     v purrr   0.3.0
## v tibble  2.0.1     v dplyr   0.7.8
## v tidyr   0.8.2     v stringr 1.4.0
## v readr   1.3.1     v forcats 0.3.0
## -- Conflicts ------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x dplyr::recode() masks car::recode()
## x dplyr::select() masks MASS::select()
## x purrr::some()   masks car::some()
```

(I use `MASS` later, but you probably won't.)

1. Work through problems 19.7 and 19.8 in PASIAS.

2. People who undergo corrective knee surgery have to do physical therapy to get themselves accustomed to walking, balancing etc. on the knee that underwent surgery. What leads people to need less time to complete their physical therapy? One researcher suggests that it is the level of physical fitness (before the knee surgery) that matters. Twenty-four male subjects ranging in age from 18 to 30 years, who had undergone corrective knee surgery within the last year, were selected for the study. The number of days required to complete the physical therapy, age, and prior level of physical fitness (categorized as low, medium or high) were recorded for each subject. The data are in `https://www.utsc.utoronto.ca/~butler/d29/rehab.txt`, along with another column `w` that we ignore. (This numbers subjects within each fitness-level group.) The data values are aligned in columns.

   (a) (2 marks) Read in the data set and display some of it.

   (b) (3 marks) Make a suitable plot that displays any relationship between age and rehab days, accounting for each subject's level of fitness. Add regression lines for each fitness level to your plot.

   (c) (3 marks) Describe briefly what your plot suggests about effects of age, fitness level and interaction.

   (d) (4 marks) Fit an analysis of covariance model predicting rehab days from the other variables, including an interaction. Display the output from `drop1`, and explain briefly what you conclude from it.

   (e) (3 marks) Explain briefly whether or not we can remove the interaction. If appropriate, fit a model without it, and test `age` and `fitness` (as a whole) for significance. What do you conclude?

(f) (3 marks) Look at the `summary` of your preferred model, and explain briefly how the conclusions from there are consistent with your graph.

3. Work through Chapter 20 of PASIAS.

4. The hook-billed kite is a bird of prey, of the same family as the eagles, which can be found in Texas, Central America, and parts of South America. One question of interest is whether males and females are of similar sizes, and if they differ in size, how.

   A number of hook-billed kites were trapped, their tail and wing lengths measured (in millimetres), and whether each bird was male or female. The kites were then released. The data are in `https://www.utsc.utoronto.ca/~butler/d29/kite.csv`.

   (a) (2 marks) Read in and display (some of) the data. What variables do you have?

   (b) (2 marks) Explain briefly why MANOVA would be a sensible method of analysis to tackle the question of interest.

   (c) (3 marks) Make a suitable graph that shows all three variables.

   (d) (2 marks) Describe briefly what you see in your plot. (I have three things in mind, and am asking you to find two of them.)

   (e) (3 marks) Create a response variable for a suitable MANOVA, and display its first few lines using `head`. Explain briefly why we need to use `head` here.

   (f) (3 marks) Run a suitable MANOVA to compare wing length and tail length for males and females. Display your results.

   (g) (2 marks) What do you conclude from your MANOVA? Explain briefly.

5. Work through Chapter 21 of PASIAS.

6. One of the supposed benefits of eating margarine rather than butter is that it reduces your cholesterol levels. In one study, 18 subjects were randomly assigned to use one of three margarines instead of butter in their diet for eight weeks. The margarines were labelled A, B, and C. Each subject had their cholesterol level measured three times: once before the study started, once after they had been using the assigned margarine for four weeks, and once after eight weeks, at the end of the study. The data are in `https://www.utsc.utoronto.ca/~butler/d29/cholesterol.csv`.

   (a) (2 marks) Read in and display (some of) the data. Convince yourself (not graded) that you have something that identifies the subjects, which margarine each subject was assigned, and measurements of cholesterol at three time points.

   (b) (5 marks) Run a suitable repeated-measures ANOVA on these data. There are some things you need to set up:

   - create a response matrix out of the appropriate columns of your data frame
   - fit a linear model with `lm` predicting your response from your between-subject factor(s)
   - get hold of a list of the different time points (eg. from your response matrix)
   - make a `data.frame` of this
   - run `Manova` suitably, displaying the result.

   (c) (3 marks) What do you conclude from your repeated-measures ANOVA? (Remember, in these you are allowed to analyze the main effects if the interaction is not significant.)

   (d) (2 marks) We are going to make a spaghetti plot to understand the significance and non-significance of the tests in the repeated-measures ANOVA. For that, we need the data in "long format" with one column of cholesterol measurements and a second column saying which time point each one comes from. Create this long-format data frame.

   (e) (3 marks) Use your long data frame to make a spaghetti plot with one spaghetti strand per subject, coloured by which margarine they were assigned to.

   (f) (3 marks) Explain briefly what you conclude from your spaghetti plot and how it matches up with what you found from your repeated measures ANOVA.

Notes

[1]My first go at this mistakenly had `%<>%` instead of a pipe, which means something (so it runs), but not what I wanted. See `https://r4ds.had.co.nz/pipes.html#other-tools-from-magrittr` for what it does.

[2]The usual: to see what it does, run it one piece at a time.

[3]If they were *females*, there would be nothing unusual about them.

[4]You may recognize this as the trick from C32 for labelling some of the points on a graph.

[5]Eventually, I will write this up properly for the lecture notes, but it's not there yet, so you are spared doing it yourself for now.

[6]Darts players like me know that triple 18 is 54 without having to think about it.